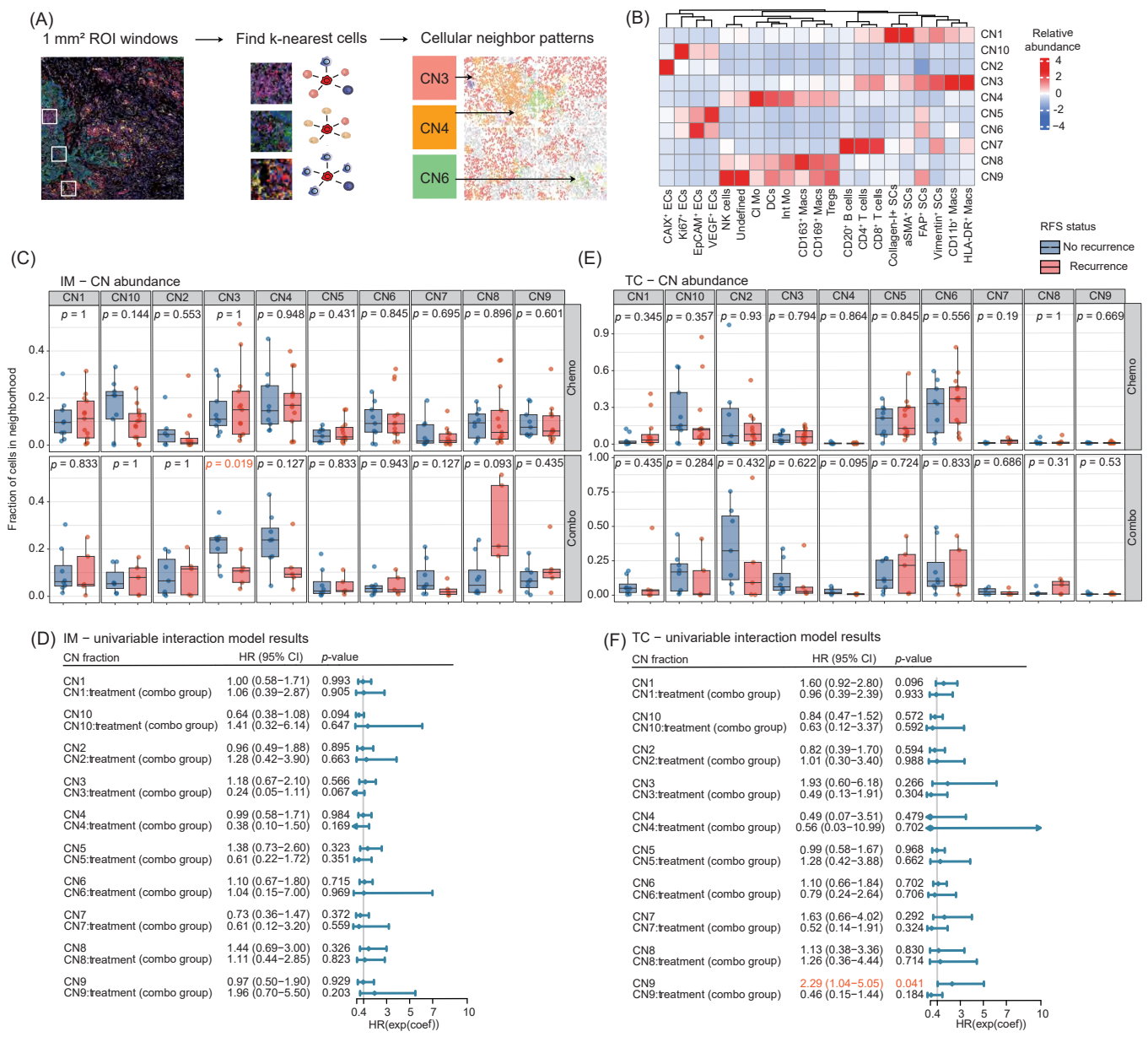

(A)
1 mm² ROI windows
Find k-nearest cells
Cellular neighbor patterns
CN3
CN4
CN6
(B)
Relative abundance
4
2
0
-2
-4
CN1
CN10
CN2
CN3
CN4
CN5
CN6
CN7
CN8
CN9
CAIX+ ECs
Ki67+ ECs
EpCAM+ ECs
VEGF+ ECs
NK cells
Undefined
Cl Mo
DCs
Int Mo
CD163+ Macs
CD169+ Macs
Tregs
CD20+ B cells
CD4+ T cells
CD8+ T cells
Collagen-I+ SCs
aSMA+ SCs
FAP+ SCs
Vimentin+ SCs
CD11b+ Macs
HLA-DR+ Macs
RFS status
No recurrence
Recurrence
(C)
IM − CN abundance
Fraction of cells in neighborhood
Chemo
Combo
(E)
TC − CN abundance
(D)
IM − univariable interaction model results
CN fraction
HR (95% CI)
p-value
HR(exp(coef))
(F)
TC − univariable interaction model results